

HYDROGENATED NBR: CATALYST-ASSISTED HYDROGENATION, BLEND FUNCTIONALIZATION, AND RESIDUAL METAL REMOVAL

1. ABSTRACT

The patent set spans the synthesis, post-treatment, and application of hydrogenated nitrile rubber (HNBR) derived from nitrile rubber (NBR), with emphasis on controlling residual unsaturation, molecular weight, and catalyst residues. The primary synthesis-focused disclosures are centered on simultaneous hydrogenation/metathesis of NBR using ruthenium- or osmium-based carbene complexes (EP2473281B1, US9150669B2), which aim to produce low-molecular-weight HNBR with narrower molecular-weight distributions and low residual double bond (RDB) content in a single step. These patents describe HNBR targets in the range of roughly 20,000 to 250,000 Mw, Mooney viscosity about 1 to 50, and MWD below 3.0, and they compare catalyst systems such as Hoveyda, Arlt, and Wilkinson-type catalysts, with catalyst loading reported in phr and metal content around 65.6 ppm in the examples. A second synthesis-oriented disclosure (EP2519350B1) is not a direct HNBR preparation patent but is relevant because it provides highly active ruthenium metathesis catalysts and explicitly mentions depolymerization/hydrogenation of NBR/HNBR under high-pressure hydrogen, with temperature windows of 30–100 °C for depolymerization and 60–150 °C for hydrogenation. The remaining patents focus on downstream formulation and purification: US9321910B2 and EP2580260A1 use HNBR in vulcanizable blends with silicone rubber or microgels to improve high- and low-temperature mechanical performance, while EP3348582B1 uses partially hydrogenated NBR as a dispersant for carbon nanotubes, specifying RDB 0.5–40 wt% and acrylonitrile content 20–50 wt%. US20190119411A1 and US7846995B2 address catalyst-removal from hydrogenated NBR solutions, using chelation, semi-coagulation, washing, and ion-exchange approaches to reduce palladium or ruthenium residues. Across the landscape, the recurring themes are selective hydrogenation of diene units, preservation of nitrile functionality, control of residual unsaturation for property tuning, and post-hydrogenation purification to enable high-performance elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example a: - Example Extracted Example Example 1::

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example a: - Example Extracted Example Example 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: EP2580260A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US20130303700A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9: - Example Extracted Example Example 10: - Example Extracted Example Example 11: - Example Extracted Example Example 12: - Example Extracted Example Example 13: - Example Extracted Example Example 14: - Example Extracted Example Example 15: - Example Extracted Example Example 16: - Example Extracted Example Example 17: - Example Extracted Example Example 18: - Example Extracted Example Example 19: - Example Extracted Example Example 20:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP3348582B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example

Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example
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 Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example
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 Comparative Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US20190119411A1
- Patent Title: Process for recovery of residual hydrogenation catalyst from hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted
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 Example 11: - Example Extracted Example Example 10: - Example Extracted Example Example 9:
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 Extracted Example Example 20:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2A: - Example Extracted Example Example 2: - Example Extracted Example Example 2:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473281B1 and US9150669B2 are closely aligned: both claim simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a Ru/Os carbene-type

catalyst, targeting low-Mw HNBR with low RDB and narrow MWD; the examples compare Hoveyda, Arlt, and Wilkinson-type catalysts, with catalyst amounts reported in phr and metal content around 65.6 ppm.

- Both EP2473281B1 and US9150669B2 emphasize single-step processing rather than the older two-step metathesis-then-hydrogenation route, and both explicitly note that the process can include a co-olefin; the extracted evidence does not reliably disclose pressure values, but the patents clearly frame the reaction as hydrogen-involved and catalyst-mediated.
- EP2519350B1 differs from the two simultaneous hydrogenation/metathesis patents by focusing on new Ru complexes for metathesis selectivity and by describing NBR depolymerization followed by hydrogenation, or hydrogenation followed by catalyst addition, at 30–100 °C for depolymerization and 60–150 °C for hydrogenation; this is a more modular route than the one-step approach in EP2473281B1/US9150669B2.
- US9321910B2 and EP2580260A1 are formulation patents rather than synthesis patents: they use HNBR or partially hydrogenated NBR as a base polymer in blends, with US9321910B2 specifying 95–100% hydrogenation of diene units and EP2580260A1 specifying partially hydrogenated NBR with about 50–99% hydrogenation; both seek improved heat/cold resistance, but via compounding rather than polymerization.
- EP3348582B1 uses partially hydrogenated NBR as a functional dispersant for carbon nanotubes and is the only primary patent in the set that explicitly ties RDB control to dispersion performance; it specifies RDB 0.5–40 wt% and acrylonitrile-derived units at 20–50 wt%, showing a broader residual-unsaturation window than the highly hydrogenated vulcanization-focused patents.
- US20190119411A1 and US7846995B2 are purification-oriented: US20190119411A1 removes residual hydrogenation catalyst from HNBR solutions by chelation plus semi-coagulation/washing, while US7846995B2 uses functionalized ion-exchange resins to remove ruthenium-containing residues from optionally hydrogenated nitrile rubber; both address catalyst contamination after hydrogenation rather than the hydrogenation step itself.
- Historically, the older patents in the set (US7846995B2, EP2473281B1/US9150669B2) emphasize catalyst removal or simultaneous hydrogenation/metathesis, whereas the newer patents (EP2580260A1, US9321910B2, EP3348582B1, US20190119411A1) increasingly use HNBR as a platform material for blends, dispersions, and cleaner downstream processing.
- Across the primary patents, the recurring process pattern is: NBR precursor with controlled acrylonitrile content, selective hydrogenation of diene units, and post-treatment to manage either molecular weight (via metathesis) or residual catalyst (via chelation/ion exchange); the main differences are whether the invention targets synthesis, compounding, or purification.
- The evidence most strongly supports low-Mw HNBR synthesis via Ru-catalyzed simultaneous metathesis/hydrogenation for the synthesis patents, while the formulation patents favor highly hydrogenated or partially hydrogenated grades depending on whether the target is vulcanizate durability or interfacial/dispersion performance.

4. CONCLUSION

The evidence shows three dominant HNBR technology directions: catalyst-enabled synthesis, property-driven compounding, and residual-metal purification. For synthesis, the strongest support is for Ru/Os carbene-catalyzed simultaneous metathesis/hydrogenation of NBR, which can reduce molecular weight while maintaining nitrile functionality and lowering residual double bonds. The most defensible synthesis target from the patent set is an HNBR with low RDB, Mw roughly 20,000–250,000, and Mooney viscosity in the 1–50 range when low-molecular-weight material is desired. For high-performance elastomer applications, the formulation patents favor highly hydrogenated HNBR, typically 95–100% hydrogenation of diene units, combined with peroxide, unsaturated carboxylic acids/salts, microgels, or silicone rubber to improve heat resistance, compression set, and low-temperature flexibility. For dispersion applications, partially hydrogenated NBR with controlled RDB and acrylonitrile content is preferred. Overall, the best-supported synthesis strategy is selective hydrogenation of NBR in solution or via simultaneous hydrogenation/metathesis under Ru-based catalysis, followed by rigorous catalyst-removal using chelation/semi-coagulation or functionalized ion exchange to obtain clean HNBR suitable for demanding automotive, sealing, and electrical applications.

5. REFERENCES

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
- EP2580260A1 | Blends of partially hydrogenated nitrile rubber and silicone rubber, ... | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2580260A1/en>
- US20130303700A1 | Blends of partially hydrogenated nitrile rubber and silicone rubber, ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US20130303700A1/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP3348582B1 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3348582B1/en>
- US20190119411A1 | Process for recovery of residual hydrogenation catalyst from hydrogenated ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US20190119411A1/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>